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A practically oriented, efficient alternative to the Monte Carlo method for measurement uncertainty estimation

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Abstract

This paper addresses the problem of uncertainty estimation for measurements made with computationally expensive numerical models, which are increasingly applied today. Typically, the Monte Carlo method (MCM) is used for uncertainty propagation in these cases. However, due to its stochastic nature, the MCM is inefficient, requiring a large number of model evaluations and thus resulting in long computation times. In this paper, an efficient alternative to the MCM is introduced. By using systematic samples of the influence quantities which are individually propagated through the measurement model, the overall number of required model evaluations can be drastically reduced. Additionally, the new method offers the possibility to simplify the measurement model for further reduced computational effort. The distributions of the propagated samples are reconstructed using a method specifically designed for systematic samples and combined to the uncertainty distribution of the measurand using convolution. The new method does not only offer increased computational efficiency but also a high information yield, as it estimates the uncertainty distribution of the measurand and the uncertainty contribution of individual influence quantities.

Keywords: measurement uncertainty, Monte Carlo method, GUM

1. Introduction

Measurement uncertainty estimation is an important task for measurements for assessing the result's reliability. To support the estimation of measurement uncertainties in practice and to improve the comparability of estimated measurement uncertainties, the Guide to the Expression of Uncertainty in Measurement (GUM) [1] has been published. For uncertainty estimation according to the GUM, a model that describes the

relationship between the influence quantities X_i of a measurement and the measurand Y is set up:

$$Y = f(X_1, X_2, \dots, X_k). \quad (1)$$

The classical GUM approach partially derives the model to calculate the so-called sensitivity coefficients $c_i = \partial f / \partial X_i$. Based on the uncertainties of the influence quantities $u(X_i)$, and the sensitivity coefficients, the combined standard uncertainty $u_c(Y)$ of the measurand can be estimated using the law of propagation of uncertainties, provided that the influence quantities are mutually uncorrelated:

$$u_c(Y) = \sqrt{\sum_i c_i^2 * u^2(X_i)}. \quad (2)$$

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While the classical GUM is an easy and practically oriented approach, it is typically only applied for relatively uncomplicated measurement models that can be expressed as a simple algebraic equation. However, with increased complexity of measurement instruments through e.g. the inclusion of advanced data evaluation software, it is difficult or even impossible to use an algebraic equation to describe measurement models. Alternatively, these models are described using a numerical form and are frequently related to as virtual instrument [2, 3]. As an analytical derivation of these models can be hard or impossible, the propagation of uncertainty contributions is typically performed numerically, using the Monte Carlo method (MCM) [4, 5].

The MCM generates a number of N_{MCM} random samples from the probability density function (PDF) of each influence quantity in the measurement model. The obtained pairs of samples are then propagated through the numerical model to calculate N_{MCM} possible values of the measurand, from which the measurement uncertainty can be estimated.

While the MCM is conceptually simple and theoretically applicable to almost all uncertainty estimation problems, it has the disadvantage of low computational efficiency. This is due to the mechanism of randomistic sampling of the influence quantities in the MCM. Typically, a large number of simulation runs, i.e. $N_{\text{MCM}} > 10\,000$ by rule of thumb, is needed to achieve satisfying calculation accuracies in the MCM. For practical models e.g. in 3D coordinate metrology using coordinate measurement machines [2] or computed tomography, where a single model evaluation can take a minute, an hour or even (much) more, this becomes a severe problem. In such cases the MCM could result in total computing time in the order of weeks and more, being completely unfeasible for practical applications.

To solve this problem, this paper presents an efficient, practically oriented alternative to the MCM. The proposed method combines the numerical uncertainty propagation of the MCM with key simplifications and assumptions from the classical GUM, such as the additivity and independence of influence quantities. Unlike the MCM however, it replaces random sampling with a systematic propagation approach, significantly improving efficiency.

The concept of the new uncertainty propagation method is presented in section 2. In section 3, the performance of the method is demonstrated using two examples. In section 4, strengths and shortcomings of the method are discussed. Section 5 presents a short summary of the new method.

2. Concept of the new method

Considering a general measurement task as described already in equation (1). It has k influence quantities $X_1, X_2, \dots, X_K$. For each influence quantity X_i ($i = 1 \dots K$), the expected uncertainty distribution represented by a PDF and its expected value $E(X_i)$ is determined via, e.g. previous experiments or specifications. In most practical cases, where X_i is obtained from direct measurements, a symmetric PDF is constructed around

the measured value of X_i , making $E(X_i)$ equal to the measured value. The core task of the measurement uncertainty analysis is to study how the uncertainty of influence quantities X_i ($i = 1 \dots k$) is propagated via the measurement system to the uncertainty of the measured value y of the measurand Y . Here, the measurand Y may have either a single output value (Y is a scalar) or multiple outputs (Y is a vector). Although the method presented in this paper is potentially applicable for both cases, we focus on the case of single output in this paper for clarity.

The schematic workflow of the new method with its deterministic sample-based uncertainty propagation is shown in figure 1. It is realized in several steps. The first step is the sampling of influence quantities based on their PDF. Noteworthy, we applied a systematic sampling (SS) approach (detailed in 2.1) rather than a random sampling method as applied in MCM. It offers a significant advantage that the sampling number can be dramatically reduced (approx. by one order), thus for reducing simulation runs and thus computing time. The j th sampling of X_i is denoted as $x_{i,j}$, where $j = 1 \dots n_i$, and n_i is the number of samples for X_i .

In the 2nd step the samples of the influence quantities $x_{i,j}$ are separately propagated through the measurement model M . Here the separated propagation of $x_{i,j}$ is understood as a model evaluation where the other influence quantities are kept constant at their expected values:

$$y_{i,j} = f(E(X_1), \dots, E(X_{i-1}), x_{i,j}, E(X_{i+1}), \dots, E(X_K)). \quad (3)$$

This setup enables the use of simplified models, as it allows certain intermediate results of the original main model to be reused during the propagation of individual influence quantities. Since the values of the other influence quantities are kept constant at their expected values, it is often unnecessary to perform a full recalculation of the complete main model for each sample. Instead, computational shortcuts can be applied by precomputing components of the model that remain unaffected by the propagated influence quantity. This can significantly reduce the overall computational effort, particularly for complicated measurement models with high computational demands.

Additionally to the propagation of the individual influence quantities samples, a reference value R is determined by setting the values of the X_i as their expected values $E(X_i)$:

$$R = f(E(X_1), \dots, E(X_{i-1}), E(X_i), E(X_{i+1}), \dots, E(X_K)). \quad (4)$$

This reference value is analogous to the estimate of the measurand obtained in the classical GUM by substituting the expected values of the influence quantities into the measurement model. However, as the proposed method accounts for potential non-linearities in the relationships between individual influence quantities and the measurand, the reference value serves only as a preliminary estimate of the measurand value. Further steps in the systematic sampling propagation reconstruction and convolution (SSPRC) method refine this estimate while incorporating these non-linear effects.

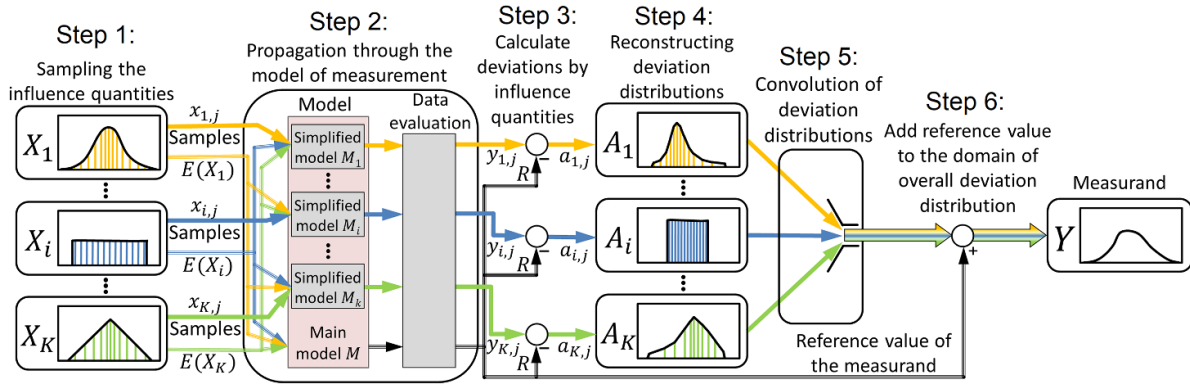


Figure 1. Schematic workflow of the SSPRC method. The influence quantities X_i are systematically sampled based on their probability density functions. Each sample is propagated individually through the measurement model, while other influence quantities are held at their expected values $E(X_i)$. Simplified models may be applied during this propagation for each influence quantity, as the separation of influence quantities allows for the reuse of intermediate results possibly avoiding some unnecessary calculations. The deviations $a_{i,j}$ are then calculated as the difference between the propagated values $y_{i,j}$ and the reference value R (the model output using $E(X_i)$ for all i). The distributions of these deviations A_i are reconstructed and combined into an overall deviation distribution using convolution. This overall deviation distribution, together with the reference value R , yields the estimated uncertainty distribution of the measurand.

In the 3rd step, the deviations $a_{i,j}$ from the reference value R , caused by the uncertainty of each influence quantity, are calculated. These deviations represent the difference between the propagated values $y_{i,j}$ and the reference value R :

$$a_{i,j} = y_{i,j} - R. \quad (5)$$

In the 4th step, the distributions of the deviation quantities A_i are reconstructed based on the $a_{i,j}$. In the 5th step, these distributions are combined into an overall deviation distribution using convolution [6]. This approach reflects the implicit assumption that the contributions of the individual influence quantities are additive, which follows from the mutual independence of these quantities. Mutual independence ensures that the combined effect on the measurand can be represented as the sum of the individual contributions. Since the influence quantities do not interact with each other, convolution is the appropriate tool to determine the resulting distribution of the summed deviations. The obtained overall deviation distribution depicts the possibly expected deviations of measurement results from the reference value due to the measurement uncertainty. To finally obtain the uncertainty distribution of the measurement, the reference value is added to the domain of the overall deviation distribution in the 6th step. It is important to note that the expected value of the final uncertainty distribution might not match the reference value. This can occur because non-linearities in the measurement model can cause the expected values of the individual deviation distributions to deviate from zero.

Based on the described workflow, the new method is referred to as SSPRC here. Like the MCM, this new method is applicable for measurements where e.g. complicated and implicit numerical algorithms are applied in the measurement model.

Noteworthy, in the development of this novel approach we do NOT aim to enhance the accuracy of the measurement uncertainty estimation, but to reduce computation loads which

is the most significant issue for complex metrology task nowadays. Compared to the state-of-the-art MCM methods, we highlight three design concepts of SSPRC for achieving this purpose. The 1st concept is the application of systematic and deterministic sampling compared to the randomistic sampling in the MCM, offering the advantage in significantly reducing the sampling number of influence quantities while their PDF still being well represented.

The 2nd concept is the separated propagation of influence quantities, where the total number of simulation runs is $N_s = \sum_{i=1}^K n_i$. While the number of model evaluations increases with the number of influence quantities, this separation enables the combination of their uncertainty contributions using convolution. As a result, it is no longer necessary, as in the MCM, to evaluate the model for a large number of combinations of all influence quantities, to combine the uncertainty contributions in the uncertainty of the measurement. This aspect is key to enable a reduction in the total number of model evaluations required by the SSPRC method. However, this concept has a cost: its applicability to manage correlated influence quantities remains limited, as these cannot be properly handled using separate propagation through the model. This will be further discussed in section 4.

The 3rd concept lies at software design practice. In the simulation of the uncertainty propagation of individual influences X_i using the SSPRC, other influence quantities are fixed at their expectation values. This fact offers an advantageous possibility that in the computation software some intermediate results can be reused to replace repeated computation procedures. This potentially leads to a further reduction in computation effort.

The individual steps of the SSPRC method are described in more detail below. However, basic steps such as setting up the numerical simulation model/software are not addressed, as they are similar for the MCM and readers can refer to related literature [1, 7–9].

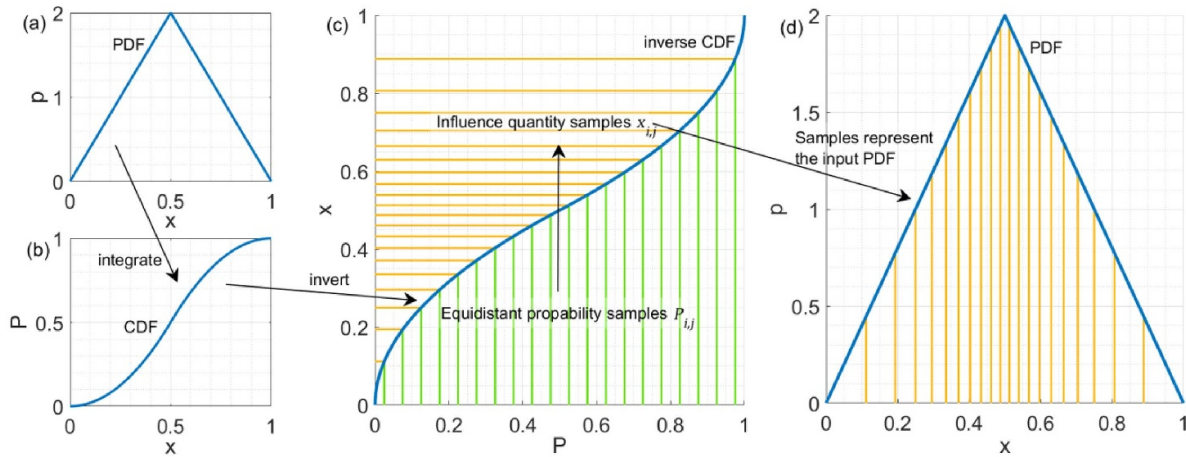


Figure 2. Principle of systematic sampling: an input PDF (a) is integrated to its CDF (b) and inverted to obtain its inverse CDF (c). This is then sampled for equidistant probability samples P_{ij} to obtain systematic samples x_{ij} of the input PDF. These samples are an accurate representation of the initial input PDF (d).

2.1. Sampling of influence quantities

The first step of the SSPRC method is to generate the samples for each influence quantity X_i . The objective thereby is to represent the PDF of every X_i as good as possible with as few samples $x_{i,j=1:N_i}$ as possible. This objective is fulfilled by a procedure which is referred to as SS. The SS is based on the Descriptive Sampling technique presented by Saliby in [10], utilizing the so called inverse cumulative distribution function (CDF) method [11] for creating the samples.

The workflow of the SS is schematically shown in figure 2. First, the PDF of an influence quantity X_i is integrated to its CDF as shown in figure 2(a). Then the CDF is inverted, leading to an inverse CDF in figure 2(b). As shown in figure 2(c), the inverse CDF is then evaluated for uniformly distributed, equidistant cumulative probabilities P_{ij} calculated from equation (6),

$$P_{ij} = (j - 0.5) / N_i \quad \text{for } j = 1 \dots N_i \quad (6)$$

where, N_i is the number of samples to be generated for X_i , and j represents the index of the sample. The samples $x_{i,j}$ generated by this procedure are a good representation of the PDF of X_i , as illustrated in figure 2(d).

Unlike for the MCM where all influence quantities are sampled with the same number N_{MCM} , for the SSPRC method individual influence quantities X_i can be sampled with an individual sampling number N_i . This offers the potential to flexibly select a proper N_i for individual X_i to have a better trade-off between the simulation accuracy and computational effort.

After the samples $x_{i,j}$ have been generated, they are propagated through the measurement model separately. For instance, for the propagation of X_k , its sampled values $x_{i=k,j}$ are input to the model, while the values of $X_{i \neq k}$ are kept constant at their expected values. Afterwards the calculation of the deviation values $a_{i,j}$ is performed as shown in figure 1.

2.1.1. Reconstruction of propagated distributions.

In the next step the distributions of the deviation quantities A_i are reconstructed from the deviation values $a_{i,j=1:N_i}$. As the distribution types of the A_i are usually unknown due to the propagation through the measurement model, flexible reconstruction methods must be used. Common methods for this purpose are the histogram method and kernel density estimation (KDE) [12].

The histogram method is a widely used approach, where the number of samples sorted into discrete bins is counted. The histogram method is principally simple. However the selection of the bin size and location affects the reconstruction performance considerably. Furthermore, the histogram method is inefficient as it requires a large number of samples to approximate a distribution well.

A generally more efficient approach is the KDE, where a reconstructed PDF is expressed as:

$$p_i(a) = \frac{1}{N_i h} \sum_{j=1}^{N_i} K\left(\frac{a - a_{i,j}}{h}\right) \quad (7)$$

where, the meaning of i, j and N_i is the same as in equation (3). $K(x)$ represents a probability kernel function, which satisfies the condition $\int_{-\infty}^{\infty} K(x) dx = 1$ and the parameter h denotes the selectable bandwidth of the kernel.

The KDE often results in better reconstructions than the histogram method, however its performance depends on a suitable choice of the kernel (e.g. Gaussian kernel [12], Epanechnikov kernel [13], etc) and the bandwidth h .

Principally both standard reconstruction methods could be used for the SSPRC, however they are not optimal for this purpose. Thus, we propose a reconstruction approach specifically designed for the systematic setup of the SSPRC, denoted as distance density estimation (DDE). The DDE uses the idea that, especially for systematically created samples, the local

density of the underlying distribution is not only encoded in the frequency of samples in a certain interval, but also by the distance of neighboring samples. The closer two neighboring deviation values $a_{i=k,j=1:N_k}$ are, the greater should be the probability density of an $A_{i=k}$ between these values.

The detailed proceeding of the DDE is as following: At first, it is verified that all samples $a_{k,j=1:N_k}$ of A_k are unique and thus a neighboring sample distance of 0 does not occur. To ensure this, all repetitive $a_{k,j}$ are combined into a single sample using a frequency parameter $f_{k,j}$ to indicate the samples repetition number. Consequently, only $N'_k \leq N_k$ unique samples remain. These are then sorted in ascending order, so that $a_{k,j} < a_{k,j+1}$. For each $a_{k,j}$ a local density $d_k(a_{k,j})$ is determined on the basis of $f_{k,j}$ and the distance between its neighboring samples as:

$$d_k(a_{k,j}) = \begin{cases} \frac{f_{k,j}}{2(a_{k,j+1} - a_{k,j})} & j = 1 \\ \frac{f_{k,j}}{a_{k,j+1} - a_{k,j-1}} & j \in [2, 3, \dots, N'_k - 1] \\ \frac{f_{k,j}}{2(a_{k,j} - a_{k,j-1})} & j = N'_k \end{cases} \quad (8)$$

Using a piece wise linear approximation, the discrete function $d_k(a_{k,j})$ is transformed into a continuous function $d_k(a)$, which is then normalized to obtain the reconstructed PDF:

$$p_k(a) = \frac{d_k(a)}{\int_{-\infty}^{\infty} d_k(a) da}. \quad (9)$$

Figure 3 illustrates the principle of the three reconstruction methods for systematic samples generated from a triangular distribution. For visualization purposes only 11 samples are used for the illustration. While the histogram method suffers from a discretization effect, the KDE provides quite a good reconstruction, although it overestimates the tails of the distribution at the cost of its central part. Compared to that, the DDE reconstructs the central part of the distribution well but shows a boundary effect at the tails. However, the strength of the DDE is, that this boundary effect at the tails reduces fast with increasing number of samples. To prove this, an experiment is carried out. In this experiment, a quantity X with a normal distribution $p_{\text{ref}}(x)$ is chosen as a reference. This X is sampled using the SS with varying sampling numbers from 10 to 1000 000. The sampled distribution of X is then reconstructed using the three reconstruction methods to obtain $p(x)$. Finally, the differences between $p(x)$ and $p_{\text{ref}}(x)$, are calculated at 100 equidistant locations x_c in a selected range from -4σ to 4σ , and consequently a performance indicator, i.e. the root-mean-square difference (RMSD) between $p(x_c)$ and $p_{\text{ref}}(x_c)$, is calculated as:

$$\text{RMSD} = \sqrt{\frac{1}{100} \sum_{c=1}^{100} (p(x_c) - p_{\text{ref}}(x_c))^2}. \quad (10)$$

In this experiment, the bin width for the histogram method has been set using Scott's Rule [14] and the center of the histogram was set to 0, according to the expectation of the distribution. For the KDE, a normal distribution kernel was selected, and the bandwidth was, following Silverman's approach

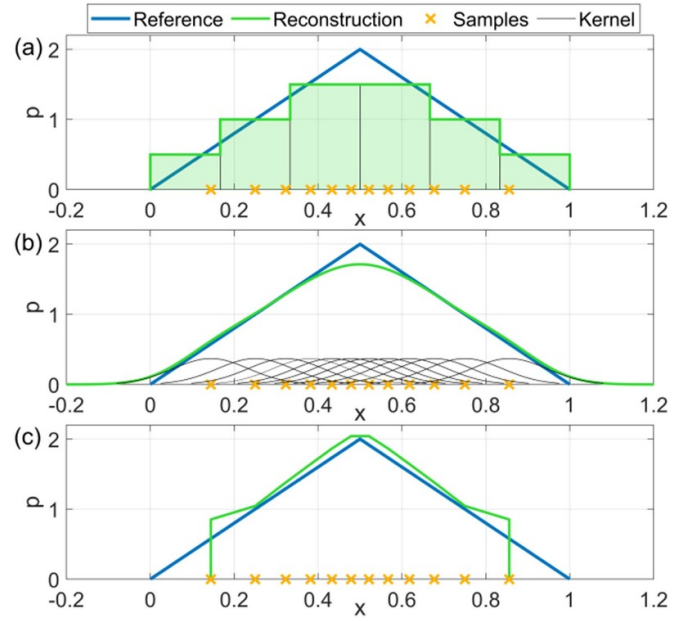


Figure 3. Reconstructed distributions from systematic samples of a triangular distribution using different methods: (a) histogram method (b) kernel density estimation (c) distance density estimation.

[12], set as, where s is the standard deviation of the generated samples.

Figure 4 shows the obtained performance metric as a function of the sample number for all three reconstruction methods. As might be expected, the largest reconstruction deviations arise for the histogram reconstruction. Compared to that, the deviations of the KDE are smaller, however, both methods have a similar convergence behavior. For up to 10 samples, the DDE has a similar performance as the KDE, however it shows a considerably faster convergence. As a result, the DDE is capable of significantly reducing reconstruction deviations from a few 10 samples onwards. While for instance the DDE results in a deviation of 3.5×10^{-3} for 100 samples, while the KDE and histogram methods need around 2000 and 40 000 samples to achieve such a small deviation, respectively.

2.2. Combining propagated uncertainty distributions

Once the individual deviation distributions of the influence quantities have been reconstructed, they are combined into an overall deviation distribution using convolution. A convolution calculates the probability distribution of the sum of two quantities based on their individual distributions. This approach of combining the individual deviation distributions is valid in the SSPRC because, through the separate propagation of the uncertainties of individual influence quantities, an implicit assumption of additivity between their contributions has been made. This additivity follows from the mutual independence of the influence quantities, which ensures that their effects on the measurand do not interact and can be summed

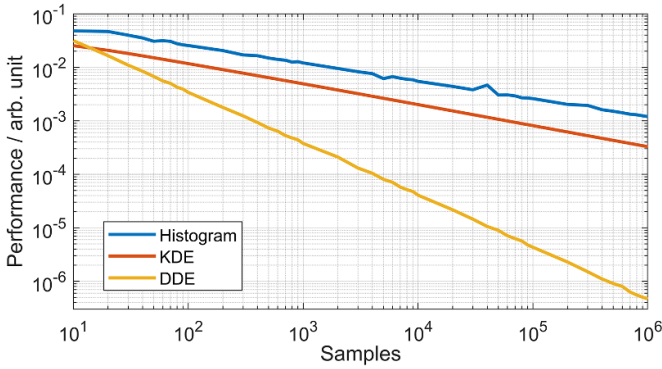


Figure 4. Reconstruction performance of a standard normal distribution from systematic samples using different reconstruction methods. A smaller performance metric stands for a better reconstruction.

directly. As an example, the convolution of two deviation distributions A_1 and A_2 can be calculated as:

$$A_o(y) = \int_{-\infty}^{\infty} A_1(x) A_2(y-x) dx. \quad (11)$$

If more than two distributions have to be combined, this can be done through a sequential application of the convolution. As the convolution can be a computationally expensive step, an efficient implementation based on the FFT method as presented in [6] can be used.

Following the calculation of the overall deviation distribution using convolution, the reference value of the measurand R , which is calculated by evaluating the main model of the measurement for the expected values of all influence quantities, is added to the domain of the overall deviation distribution to obtain the uncertainty distribution of the measurement.

3. Experiments

In this section, the basic performance and efficiency of the SSPRC method is demonstrated. For this purpose, two examples are given. For a direct comparison of the SSPRC and MCM, in the first example an intentionally designed simple and computationally inexpensive but non-linear fictive measurement model is selected, while the second example is taken from practice instead. In this second example the uncertainty of a projected area function (PAF) of a microhardness indenter measured with an atomic force microscope (AFM) is estimated. In this example, the model of the measurement is computation-intensive, which aims to demonstrate the performance of the SSPRC for real application scenarios.

3.1. Experiment 1: a simple fictive measurement

For the first experiment a simple fictive measurement with the three influence quantities X_1 , X_2 and X_3 is modeled as:

$$Y = (X_1 + X_2^2) X_3. \quad (12)$$

The uncertainty quantity X_1 is assumed to be normally distributed with an expectation value $\mu = 2$ and a standard deviation $\sigma = 0.2$. Quantity X_2 is regarded as uniformly distributed with $\mu = 0.5$ and a range $R = 0.9$ and quantity X_3 as symmetrically triangularly distributed with $\mu = 5$ and width $R = 0.6$.

At the beginning of this experiment four different measurement uncertainty estimations are carried out, using twice the MCM and twice the SSPRC. First, for calculating a reference uncertainty distribution the MCM is applied with 10^7 model evaluations. Then the MCM is applied with a significantly reduced, but for computationally expensive models still large number of 5000 model evaluations. To generate the (pseudo-) random samples of the influence quantities for the MCM, the Mersenne Twister algorithm [15] is used to create uniformly distributed samples between 0 and 1, which are then transformed into the respective distribution using the inverse CDF method [11]. Following the MCM, the SSPRC method is applied twice for uncertainty estimation. For the first application 100 samples per influence quantity are selected, to show the SSPRC's performance and efficiency for a low number of samples. After that, the SSPRC is applied using 10^7 samples per influence quantity to compare its performance with a low and high number of model evaluations.

Figure 5(a) shows the uncertainty distributions of the measurand Y obtained from above four estimations. For the MCM with 5000 samples, significant deviations to the reference distribution can be seen, indicating the detrimental influence of the (pseudo-) random sampling on the MCM's accuracy and efficiency. Compared to that, the SSPRC estimates the uncertainty distribution well with high and even with low sample numbers. These results on the one hand indicate the principal feasibility of the SSPRC for uncertainty estimation and on the other hand show its efficiency compared to the MCM.

As the results in figure 5(a) can be biased by visual impression and to compare the SSPRC and MCM more objectively, the precision of both methods with increasing number of samples, i.e. their convergence and thus their efficiency, is examined. For sample numbers between 10 and 10^6 the relative error of the standard uncertainty for Y is calculated to a reference with 10^8 samples. For the MCM, the calculations are repeated 10 000 times and the median error and the 95% confidence interval are evaluated. The results are illustrated in figure 5(b), showing a superior convergence behavior of the SSPRC method compared to MCM and large confidence intervals for the standard uncertainties determined by the MCM. If for the fictive measurement a standard uncertainty should be determined with a relative error of 1% or below, the MCM on average would need around 2000 model evaluations, while the SSPRC would need around 600 model evaluations with 200 samples per influence quantity. This concrete example again demonstrates the efficiency of the SSPRC method, even if the performance difference between both methods seems to be not specifically large. However, it must be taken into account that the number of model evaluations for the MCM only refers to the median value. Since an uncertainty estimate should principally be reproducible and objective, it is often not sufficient to reach a certain precision with a probability of only 50% as for the median case. If therefore instead the upper limit of the

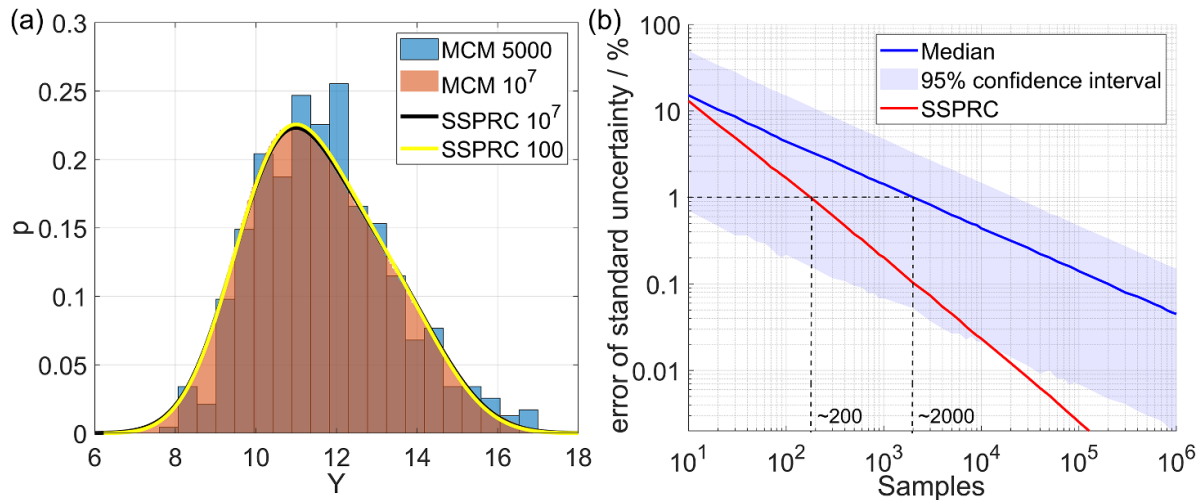


Figure 5. (a) Uncertainty distribution of the measurand from different uncertainty estimations; (b) convergence of the standard uncertainty estimated with the SSPRC method and MCM. As indicated the MCM needs approximately ten times as many samples per influence quantity to reach an error of 1% than the SSPRC.

95% confidence interval is used as the reference for the MCM, not 2000 but 20 000 model evaluations are required, clearly and undoubtedly highlighting the efficiency and reproducibility advantage of the SSPRC.

3.2. Experiment 2: computationally expensive model

To demonstrate the practical advantages of the SSPRC compared to the MCM, both methods are used in the second example to determine the measurement uncertainty of an AFM measurement. The uncertainty estimation for the considered measurement is based on a computationally expensive numerical model, where the application of the MCM results in disproportionately large computation times.

3.2.1. Measurement setup. The aim of the measurement is to obtain the so-called PAF of a microhardness indenter. The PAF is the vertically projected surface area of the indenter as a function of the height from its apex, which is one of the key parameters needed for accurate evaluation of mechanical properties of materials in indentation measurements [16].

To measure the PAF, here an AFM is applied, which sequentially scans the topography of the indenter using a sharp tip. The result of the measurement is a topography image, in which the individual pixels represent the indenter's local height. To obtain the PAF from the topography image, the number of pixels in height intervals of 0.5 nm is counted and integrated over the indenter height from the apex. Finally, the integrated number of pixels is multiplied with the projected area of each pixel, which is simply the pixel size.

The instrument used for the measurement is the metrological large range AFM (Met. LR-AFM) of the Physikalisch-Technische Bundesanstalt [17]. The Met. LR-AFM is a highly accurate and versatile tool. It is based on a nanomeasuring machine [18], providing a large scanning range of 25 mm × 25 mm × 5 mm (x, y, z). The scanning motion of the

Met. LR-AFM is traceably measured using three laser interferometers. To minimize the effect of Abbe errors, the Met. LR-AFM fulfills the Abbe principle. For minimizing environmental effects on a measurement, the Met. LR-AFM is placed on a damping stage inside an acoustically insulating chamber in a climate controlled clean room.

With the Met. LR-AFM an 8 $\mu\text{m} \times 8 \mu\text{m}$ area around the indenter apex has been measured using a scanning speed of 1 $\mu\text{m s}^{-1}$ and 1000 × 1000 pixels. Figure 6(a) shows the obtained topography image of the indenter. The PAF evaluated from this topography image is shown in figure 6(b).

3.2.2. Influence quantities and model. For the uncertainty estimation using the MCM and SSPRC the relevant influence quantities of the measurement are identified. These are the geometry of the AFM tip, the noise of the AFM, a relative drift between the AFM tip and the indenter during the measurement, and a tilted installation of the indenter in the AFM. These influence quantities have been characterized as follows:

- Based on experience the AFM tip is assumed to be spherical with a radius uniformly distributed between 2 nm and 20 nm.
- The measurement noise is characterized as normally distributed white noise with a standard deviation σ . σ has a uniform distribution between 0.2 nm and 0.3 nm.
- The drift is assumed to be up to 20 nm for each coordinate axis of the AFM during measurement. Due to a lack of detailed information about the actual drift, it is approximated as uniformly distributed with a linear temporal behavior.
- Based on observations the installation tilt of the indenter in the AFM is estimated to be as large as 1.5° in any direction from its central axis. Both, the magnitude and direction of the tilt are assumed to be uniformly distributed.

The identified influence quantities are assumed to be uncorrelated, as they arise from independent mechanisms within the

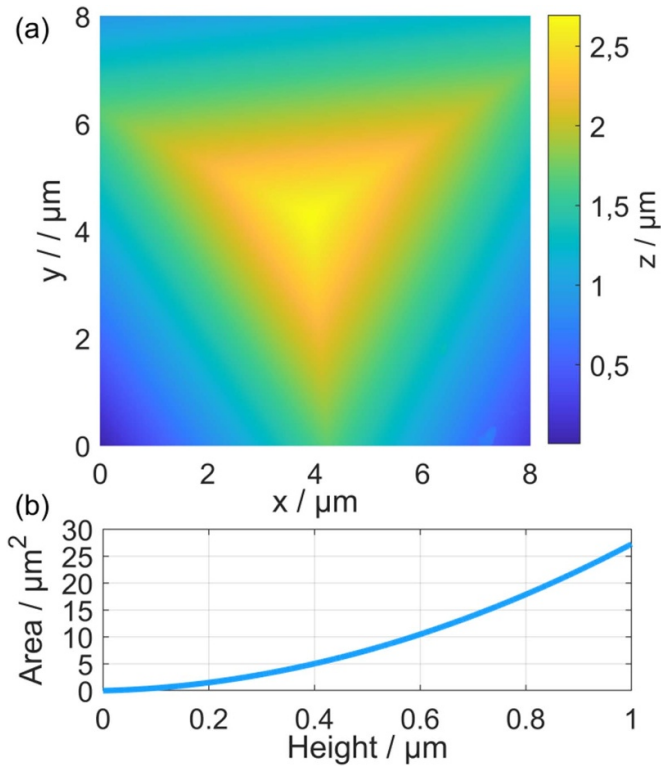


Figure 6. (a) Measured topography of the hardness indenter; (b) area function evaluated from the measured topography.

AFM. This assumption is based on the design and functionality of the AFM, allowing confident application of the SSPRC method.

Using the identified influence quantities, the measurement model is established with the following steps. First, the measurement image, used as the starting point for the simulations, is tilted to account for the installation tilt influence. This is achieved by multiplying the x , y , and z coordinates of the pixels by a rotation matrix. Afterwards the acquisition of individual pixels is simulated. For this, first the nominal measurement position of the tip is modified by the simulated drift. Subsequently, the contact point of the simulated AFM tip with the indenter is calculated using an interpolation operation to determine the locally measured sample height [19], to which then random noise is added. These steps are repeated for all pixels of the measurement image. The complete simulation workflow for this is implemented in a Matlab script.

The model described above is directly used for the MCM and serves as the main model for the SSPRC. However, for the uncertainty propagation of individual influence quantities with the SSPRC, simplified and more efficient models can be derived. In the main model, most of the computational effort arises from calculating the contact points between the AFM tip and the tilted indenter. The calculation of an indenter installation tilt disrupts the regular xy -grid of the pixels in the initial noise-filtered reference topography image, necessitating the use of the computationally inefficient ‘scattered-Interpolant’ Matlab function, which works with irregularly gridded input data. Since the expected value of the indenter

Table 1. Number of model evaluations and required computing time for uncertainty estimation of the indenter PAF.

Method	Contribution	N	Computing time
SSPRC	Noise	2000	24 s
SSPRC	Tip	100	20 s
SSPRC	Tilt	2000	~20 min
SSPRC	Drift	1331	41 s
SSPRC	Total	5431	<25 min
MCM	Total	10 000	~56 h

installation tilt is zero, its influence is omitted from the models used for the propagation of other influence quantities within the SSPRC method. This simplification is possible because, in the SSPRC, each influence quantity is propagated individually while all other quantities are fixed at their expected values. As the tilt’s expected value is zero, it does not introduce a systematic deviation in the propagated results of other influence quantities. While random variations in the tilt influence exist, these are explicitly considered when the tilt itself is propagated as an influence quantity. Consequently, for the other influence quantities the more efficient ‘griddedInterpolant’ function can be used for the contact point calculations when separately propagating their samples. This and other optimizations of the main model for the propagation of individual influence quantities result in simplified models for each influence quantity and reduced overall computation times, as is documented in the following section. For readers who wish to gain deeper insights into the computational model and the application of SSPRC or MCM, example code is available by request from the authors.

3.2.3. Results. For uncertainty estimation using the MCM, 10 000 model evaluations have been carried out as a reference. For this, a computation time of around 56 h has been required using an Intel Core i9-9900 processor. For the SSPRC the number of model evaluations for each influence quantity and the required computation time has been given in table 1. As can be seen, in total approximately 5000 model evaluations were carried out for the SSPRC, whereby the number of evaluations for each influence quantity was chosen as considered necessary for an accurate propagation of the respective uncertainty contribution.

The overall computation time needed for the SSPRC was less than 25 min, corresponding to 0.28 s per average model evaluation. In comparison to that, the MCM needed around 56 h for the 10 000 model evaluations, corresponding to 20 s per individual evaluation. This difference in evaluation time equates to an efficiency improvement of a factor 70 for the SSPRC compared to the MCM, impressively showing the SSPRC’s potential benefit by using simplified measurement models.

In order to assess whether the uncertainties determined with both methods are comparable and the reduction of computational time for the SSPRC is not at the expense of a relevant reduction in estimation accuracy, Figure 7(a) shows the determined relative standard uncertainties of the PAF for

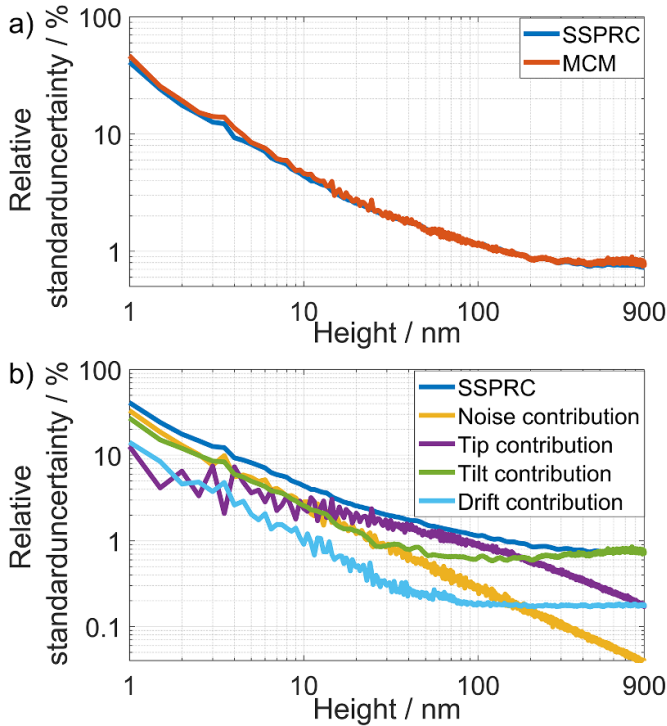


Figure 7. Relative standard uncertainty of the micro hardness indenter PAF. (a) Uncertainty determined with the SSPRC and MCM. (b) Contribution of the individual influence quantities to the measurement uncertainty determined by the SSPRC.

each method. It can be seen that both methods have estimated similar uncertainty curves over the indenter height. Only for indenter heights below 20 nm small differences between both uncertainty curves can be seen.

An aspect that differentiates the SSPRC and MCM is the SSPRC's intrinsic capability of providing an information about the uncertainty contribution of the individual influence quantities to the overall measurement uncertainty. With the SSPRC, e.g. the standard deviation of the respective reconstructed deviation distribution can be analyzed. As can be seen in figure 7(b), where the relative standard uncertainty caused by each influence quantity is plotted, such an information is practically helpful for identifying the most relevant sources of uncertainty.

Besides the determination of a relatively simple metric like a standard uncertainty, both, the MCM and SSPRC were used to estimate the PAF uncertainty distribution. For better visibility, figure 8 shows the distribution of the simulated deviation of the projected indenter surface area from the reference surface area of the indenter geometry used as starting point for the simulations at a height of 200 nm from the indenter apex. Both methods estimate principally a very similar distribution, only a practically negligible offset of both distributions can be observed. However, with the MCM, despite the use of 10 000 model evaluations, the influence of its random sampling can still be seen in the histogram representation of the distribution, while the SSPRC determines a smooth distribution.

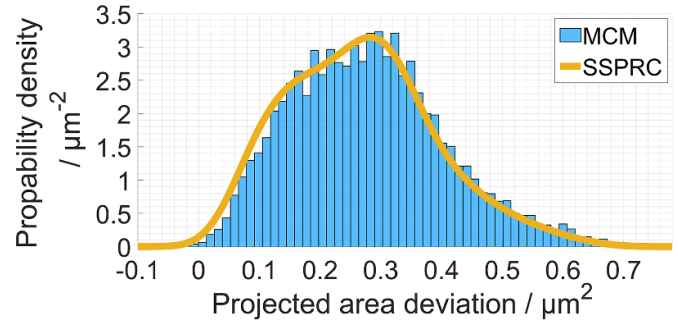


Figure 8. Distribution of the total deviations of the simulated projected indenter surface area from the projected reference surface area at a height of 200 nm from the indenter apex. The total deviations were obtained by convolving the individual deviations defined in equation (5), as described in Step 5 of the SSPRC method.

Overall, it can be concluded for this example that the MCM and SSPRC provided well agreeing results, whereby the SSPRC required significantly less computing time (about a factor of 70) and yielded more information through the estimation of uncertainty contributions of individual influence quantities.

The two application examples above have illustrated the feasibility of the developed SSPRC method for uncertainty estimation and highlighted its superior computational efficiency over the MCM. Owing to increasing applications of advanced data analytic techniques such as big data, machine learning and artificial intelligence in modern measurement tools, the algorithms and software used in measurement models are becoming much more complex today. Consequently, measurement uncertainty evaluation becomes more computation-intensive when the MCM method is applied. The new SSPRC approach proposed in this paper offers a highly potential solution for such complicated future uncertainty estimation tasks.

4. Discussion

The SSPRC method offers significant advantages over the MCM, but it also has some limitations, which are discussed below. Additionally, potential future extensions of the SSPRC and an alternative approach are presented.

The great strength of the SSPRC method is its efficiency compared to the MCM. This efficiency is achieved through two aspects. The first aspect is the systematic nature of the SSPRC. By avoiding unnecessary stochastic variations which occur in the MCM and have to be compensated by large sample numbers, the SSPRC can provide good results with significantly reduced sampling numbers. Mainly three components contribute to the systematic and efficient nature of the SSPRC. First, the SSPRC method avoids stochastic variations of the samples using its SS approach. Second, on the basis of the systematically generated and propagated samples, the

DDE provides accurate reconstructions of the deviation distributions caused by the respective influence quantities, even if only small numbers of a few ten or hundred samples are used. Finally, the combination of the propagated uncertainties by convolution again avoids stochastic variations compared to the MCM, where the uncertainty combination of the influence quantities is achieved by propagating random pairs of sample values for these quantities.

The second aspect contributing to the SSPRC's overall efficiency is, that with this method especially for simulation-based models of a measurement, the model can be simplified for simulating the error propagation of individual influence quantities. As shown in the case study of uncertainty estimation of the PAF in section 3.2, this can lead to a significant reduction in the computing time compared to the MCM.

Besides its efficiency, another major advantage of the SSPRC method is the amount of information yielded from its application. In contrast to the classical GUM, the SSPRC method directly estimates the PDF of the measurand, which is important for the statement of confidence intervals. In addition to that, the SSPRC method has the advantage over the MCM that it inherently provides information on the contribution of the individual influence quantities to the overall measurement uncertainty. This is important for identifying dominant error sources of a measurement, which is a key step for improving it.

A possible way to further improve the computational efficiency and precision of the SSPRC is to replace the DDE and the convolution used in the SSPRC for reconstructing and combining the deviation distributions with the calculation of empirical characteristic functions (ECFs) [20]. For each deviation distribution, an ECF can be calculated from the deviation samples. All ECFs can then be multiplied to obtain a combined ECF, which can subsequently be transformed into the overall deviation distribution using the inverse Fast Fourier Transform. This process is expected to be more computationally efficient, as it avoids the sequential application of convolution. However, the difference in computational efficiency is likely to be small compared to the time required for evaluating the computationally expensive measurement models for which the SSPRC is primarily designed. On the other hand, the ECF approach should also be able to provide more precise estimations of the overall deviation distribution, provided that the frequency range for creating the ECF is set appropriately. While the ECF approach offers intriguing potential and warrants further exploration, the choice of the ECF's frequency range introduces a parameter that affects performance. For this reason, the parameter-free and thus simpler and more reliable DDE method is used in this paper.

While the SSPRC offers many advantages, there are yet some limitations in its applicability. A condition for the application of the separated propagation and the combination of the influence quantities uncertainties using convolution is, that their uncertainties do not affect each other during the propagation and thus, that their effects on the measurand are additive. At first this condition appears to be quite restrictive. However, it also applies to the law of propagation of uncertainties of the

classical GUM in equation (2), which is widely used in practice with sufficient accuracy. For most practical problems it is often adequate if this condition is fulfilled approximately, which is frequently the case, as e.g. for the examples given in section 3.

Another limitation of the SSPRC is its applicability to correlated influence quantities, as these can not be handled properly using the separate propagation through the model. It is therefore important to verify if the influence quantities are uncorrelated when applying the SSPRC method. This can be achieved through diagnostic checks or a careful evaluation of the measurement procedure. A possible solution if two or more influence quantities are correlated, is to propagate the samples of the correlated influence quantities together through the model, as it is done using the MCM or better the Descriptive Sampling of Saliby [10].

A possible alternative to the SSPRC, which is able to take correlations and non-additive behavior into account, is the use of a surrogate model [21]. A surrogate model is an empirical model that approximates the behavior of the original model, but is less computationally intensive, making it suitable for use with the MCM. However, to create a surrogate model, data on the behavior of the original model has to be available, which is typically determined through corresponding evaluations of the original model for various combinations of the influence quantities, which again generates considerable computational effort. Afterwards, the surrogate model, which must have a suitable form in order to be able to reproduce the original model's behavior, is then trained on the obtained data. Due to the effort involved in generating a suitable surrogate model, this approach is only efficient, if the model can be reused multiple times. If, for example, uncertainties are to be estimated frequently for a measurement system whose behavior does not change, but only the values of the influence quantities, the use of a surrogate model can be more suitable than the SSPRC method. However, for applications such as the uncertainty determination of the PAF from section 3.2, where the geometry and thus the propagation of uncertainties differs for each indenter, the SSPRC is probably the more efficient approach.

A final aspect of the SSPRC that needs to be discussed is the selection of sample numbers for the individual influence quantities. In the indenter example in section 3.2, the number of samples for each influence quantity was chosen subjectively. However, this approach could result in selecting too few samples for some influence quantities, leading to inaccurate uncertainty estimation. Conversely, selecting too many samples would unnecessarily increase computation time. To address this issue, an adaptive sampling approach similar to the adaptive MCM [4] could be implemented for the SSPRC. In this approach, new samples would be iteratively generated and propagated through the model until the estimated uncertainty converges. While the number of newly generated samples can be chosen flexibly in the MCM, for the SSPRC, the number of samples for an influence quantity X_i would need to be increased by $N_i + 1$ to maintain the systematic structure of the samples and to place the new samples in between the

previous ones. To determine whether enough samples have been generated, the convergence of the standard deviation and the expected value of the propagated samples could be analyzed. Additionally, also the relative uncertainty contribution of the propagated samples could serve as a stopping criterion for the further sampling of an influence quantity. As an example, for influence quantities with a small observed uncertainty contribution, sampling could be stopped early, as they are only of minor importance for the overall uncertainty.

5. Summary

In this paper, an efficient alternative to the MCM for measurement uncertainty estimation has been presented. For measurements described by computationally expensive numerical models, the SSPRC method can significantly reduce computational time compared to the MCM. This is achieved by propagating the uncertainties of the influence quantities separately and systematically through the measurement model, avoiding randomistic influences in the uncertainty propagation and reducing the number of required model evaluations. Additionally, due to the separate propagation of the influence quantities' uncertainties, the SSPRC enables the use of simplified and more computationally efficient measurement models for each influence quantity, further reducing the overall computational effort.

The SSPRC method provides a high informative yield, as it can estimate both, the uncertainty distribution of the measurand and the relative uncertainty contributions of individual influence quantities. Neither the classical GUM nor the MCM offer these capabilities simultaneously.

Using two application examples, the potential of the SSPRC has been presented. Although the SSPRC has a few disadvantages, it presents a promising alternative to the MCM for computationally expensive measurement models.

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References

- [1] BIPM, IEC, IFCC, ILAC, ISO, IUPAC, IUPAP, and OIML 2008 Evaluation of measurement data Guide to the Expression of Uncertainty in Measurement vol 100 (Joint Committee for Guides in Metrology, JCGM) (available at: www.bipm.org/documents/20126/2071204/JCGM_100_2008_E.pdf/cb0ef43f-baa5-11cf-3f85-4dc86f77bd6)
- [2] Kok G, Wübbeler G and Elster C 2022 Impact of imperfect artefacts and the modulus operandi on uncertainty quantification using virtual instruments *Metrology* **2** 311–9
- [3] Balsamo A, Di Ciommo M, Mugno R, Rebaglia B I, Ricci E and Grella R 1999 Evaluation of CMM uncertainty through Monte Carlo simulations *CIRP Ann.* **48** S.425–8
- [4] BIPM, IEC, IFCC, ILAC, ISO, IUPAC, IUPAP, and OIML 2008 Evaluation of measurement data | supplement 1 to the \guide to the expression of uncertainty in measurement *Propagation of Distributions Using a Monte Carlo Method* vol 101 (Joint Committee for Guides in Metrology, JCGM) (available at: www.bipm.org/documents/20126/2071204/JCGM_101_2008_E.pdf/325dcaad-c15a-407c-1105-8b7f322d651c)
- [5] Metropolis N and Ulam S 1949 The Monte Carlo method *J. Am. Stat. Assoc.* **44** 335–41
- [6] Korczynski M J *et al* 2006 Convolution and uncertainty evaluation *Advanced Mathematical and Computational Tools in Metrology VII* (P Caparica) pp 188–95
- [7] BIPM, IEC, IFCC, ILAC, ISO, IUPAC, IUPAP, and OIML 2020 *Guide to the expression of uncertainty in measurement* Part 6: Developing and using measurement models (Joint Committee for Guides in Metrology, JCGM GUM-6) (available at: www.bipm.org/documents/20126/2071204/JCGM_GUM_6_2020.pdf/d4e77d99-3870-0908-ff37-c1b6a230a337)
- [8] Dieck R H 2007 *Measurement Uncertainty: Methods and Applications* (ISA)
- [9] Hampel B *et al* 2018 Approach to determine measurement uncertainty in complex nanosystems with multiparametric dependencies and multivariate output quantities *Meas. Sci. Technol.* **29** S.035003
- [10] Saliby E 1990 Descriptive sampling: a better approach to Monte Carlo simulation *J. Oper. Res. Soc.* **41** S.1133–42
- [11] Gentle J E 2003 *Random Number Generation and Monte Carlo Methods* vol 381 (Springer)
- [12] Silverman B W 1998 *Density Estimation for Statistics and Data Analysis* (Chapman & Hall/CRC)
- [13] Epanechnikov V A 1969 Non-parametric estimation of a multivariate probability density *Theory Probab. Appl.* **14** S.153–8
- [14] Scott D W 2010 Scott's rule *Wiley Interdiscip. Rev. Comput. Stat.* **2** S.497–502
- [15] Matsumoto M and Nishimura T 1998 Mersenne twister: a 623-dimensionally equidistributed uniform pseudo-random

- number generator *ACM Trans. Model. Comput. Simul.* **8** S.3–30
- [16] Dai G, Herrmann K and Menelao F 2009 Two approaches for enhancing the accuracy of the Rockwell hardness test *Meas. Sci. Technol.* **20** 065701
- [17] Dai G, Wolff H, Pohlenz F and Danzebrink H-U 2009 A metrological large range atomic force microscope with improved performance *Rev. Sci. Instrum.* **80** S.043702
- [18] Manske E, Hausotte T, Mastilo R, Machleidt T, Franke K H and Jäger G 2007 New applications of the nanopositioning and nanomeasuring machine by using advanced tactile and non-tactile probes *Meas. Sci. Technol.* **18** 520
- [19] Bukharaev A A *et al* 1998 Three-dimensional probe and surface reconstruction for atomic force microscopy using a deconvolution algorithm *Practice* **5** S.21
- [20] Yu J 2004 Empirical characteristic function estimation and its applications *Econ. Rev.* **23** 93–123
- [21] Queipo N V, Haftka R T, Shyy W, Goel T, Vaidyanathan R and Tucker P K 2005 Surrogate-based analysis and optimization *Prog. Aerospace Sci.* **41** 1–28